JIALE SHI

Postdoctoral Associate
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EB-1A (Extraordinary Ability) I-140 – Approved

Research Skills and Interests

Statistical Physics	Molecular Simulation	Complex Polymer and Soft Matter
Generative AI	Graph Neural Network	Large Language Models
Data Science	Applied Math	Chemical Informatics

Education & Research Experience

6/2025-present **Postdoctoral Associate, Boston University**
Advisor: **Qiang Cui**

- Developed graph models to quantify protein dynamics distance
- Modeled pH and buffer effects on lipid nanoparticle phase transitions
- Developing conditional generative models for lipid molecule design
- Developing diffusion models to accelerate the field-theoretical simulation for the liquid liquid phase separation of lipid nanoparticles for mRNA delivery

4/2022-6/2025 **Postdoctoral Associate, Massachusetts Institute of Technology (MIT)**
Advisor: **Bradley Olsen**, Co-advisor: **Debra Audus** (NIST)
Worked on **CRIPT (5 million NSF project)**, focused on developing polymer informatics and data-driven polymer property predictions

- Developed chemistry-informed functions to precisely quantify polymer similarity
- Developed MacroSimGNN to accelerate macromolecule similarity calculations
- Developed landmark distance embedding for polymers
- Developed BigFingerprint, a topology-aware fingerprint for polymers
- Developed LLM-based Knowledge Graph RAG for Polymer Simulation

2022 **Ph.D. in Chemical Engineering (GPA 4.0/4.0), University of Notre Dame**
Advisor: **Jonathan Whitmer**
PhD Thesis: Computing Free Energy Landscapes for Materials Design
Minor in Computational Science and Engineering

- Developed AI models to predict polymer-surface adhesion free energy
- Developed deep transfer learning to overcome data-scarcity challenges
- Modeled temperature-dependence of anisotropic elastic responses of liquid crystal
- Modeled temperature-dependence of dynamics and stability of cluster isomerization

2017 **B.S. in Chemistry (GPA 3.7/4.0, Top 10%), Peking University**

AWARDS & HONORS

2023 ACS PMSE Future Faculty Scholar

2023 Finalist, NIST PEAR Accolades Outstanding Technical
2023 Big Data Award at Big Data in Polymer Chemistry Session, ACS POLY
2023 Selected Attendees of ACS Postdoc to Faculty (P2F) Workshop
2023 Selected Attendees of Soft Matter Future Faculty Workshop
2023 Winner, MIT ChemE Teach-Off Competition
2023 Travel Grant of Forum for Early Career Scientists (FECS), APS
2020 Graduate School Professional Development Grant, University of Notre Dame
2020 Graduate Student Union Conference Presentation Grant, Notre Dame
2020 Travel Grant of Division of Soft Matter (DSOFT), APS
2020 Outstanding Paper Award, University of Notre Dame
2019 Graduate School Professional Development Grant, University of Notre Dame
2019 Graduate Student Union Conference Presentation Grant, Notre Dame
2019 Best Poster Award, 6th ND-Purdue Soft Matter & Polymers Symposium
2014-2017 Cyrus Tang Scholarship, Peking University
2014 National Endeavor Fellowship, Peking University
2012 2nd Prize in China's National Olympic Chemistry Competition

PUBLICATIONS

Published

13 **Jiale Shi**, Runzhong Wang, Nathan J. Rebello, Jiarui Lu, Debra J. Audus, Bradley D. Olsen. "MacroSimGNN: Efficient and Accurate Calculation of Macromolecule Pairwise Similarity via Graph Neural Network." In revision. ***Macromolecules***. 2025. *Preprint Link*: DOI: 10.26434/chemrxiv-2024-8hs2k

12 Joren Van Herck,.., **Jiale Shi**,..., Jonathan K. Whitmer,..., Berend Smit. "Assessment of Fine-Tuned Large Language Models for Real-World Chemistry and Material Science Applications." ***Chemical Science*** 2025, 16, 670-684. DOI: 10.1039/D4SC04401K

11 **Jiale Shi**, Dylan Walsh, Nathan J. Rebello, Weizhong Zou, Michael E. Deagen, Katharina A. Fransen, Xian Gao, Bradley D. Olsen, Debra J. Audus. "Calculating Pairwise Similarity of Polymer Ensembles via Earth Mover's Distance." ***ACS Polymers Au*** 2024, 4, 1, 66–76. DOI: 10.1021/acspolymersau.3c00029

10 Qianxiang Ai, Fanwang Meng, **Jiale Shi**, Brenden Pelkie, Connor W. Coley. "Extracting Structured Data from Organic Synthesis Procedures Using a Fine-Tuned Large Language Model." ***Digital Discovery*** 2024, 3, 1822-1831. DOI: 10.1039/D4DD00091A

9 Nathan J. Rebello, Akash Arora, Hidenobu Mochigase, Tzyy-Shyang Lin, **Jiale Shi**, Debra J. Audus, Eric. S. Muckley, and Bradley D. Olsen. "BCDB: The Block Copolymer Phase Behavior Database." ***Journal of Chemical***

Information and Modeling 2024, 64, 16, 6464-6476. DOI: 10.1021/acs.jcim.4c00242

8 Yoel Zimmermann,…, **Jiale Shi**,…, Ben Blaiszik. "Reflections from the 2024 Large Language Model (LLM) Hackathon for Applications in Materials Science and Chemistry---Knowledge Graph RAG for Polymer Simulation." ***Arxiv*** 2024. DOI: 10.48550/arXiv.2411.15221

7 **Jiale Shi**, Nathan J. Rebello, Dylan Walsh, Weizhong Zou, Michael Deagen, Bruno Salomao Leao, Debra J. Audus, Bradley D. Olsen. "Quantifying Pairwise Similarity for Complex Polymers." ***Macromolecules*** 2023, 56, 18, 7344-7357. DOI: 10.1021/acs.macromol.3c00761 (Polymer similarity function is designed for the search engine of Community Resource for Innovation in Polymer Technology (CRIPT) Project)

6 Kevin Maik Jablonka,..., **Jiale Shi**,..."14 Examples of How LLMs Can Transform Materials Science and Chemistry: A Reflection on a Large Language Model Hackathon." ***Digital Discovery*** 2023, 2, 1233-1250. DOI: 10.1039/D3DD00113J

5 **Jiale Shi**, Fahed Albreiki, Yamil J. Colón, Samanvaya Srivastava, Jonathan K. Whitmer. "Using Transfer Learning to Leverage Prior Knowledge in the Prediction of Adhesive Free Energies between Polymers and Surfaces." ***Journal of Chemical Theory and Computation***. 2023, 19, 14, 4631-4640. DOI: 10.1021/acs.jctc.2c01314

4 **Jiale Shi**, Michael J. Quevillon, Pedro Henrique Amorim Valença, Jonathan K. Whitmer. "Predicting Adhesive Free Energies of Polymer-Surface Interactions with Machine Learning." ***ACS Applied Materials & Interfaces*** 2022, 14, 32, 37161–37169. DOI: 10.1021/acsami.2c08891

3 **Jiale Shi**, Shanghui Huang, François Gygi, Jonathan K. Whitmer. "Free Energy Landscape and Isomerization Rates of Au_4 Clusters at Finite Temperature." ***The Journal of Physical Chemistry A*** 2022, 126, 21, 3392-3400. DOI: 10.1021/acs.jpca.2c02732

2 **Jiale Shi***, Hythem Sidky*, Jonathan K. Whitmer. "Automated determination of n-cyanobiphenyl and n-cyanobiphenyl binary mixtures elastic constants in the nematic phase from molecular simulation." ***Molecular Systems Design & Engineering*** 2020, 5, 1131-1136. DOI: 10.1039/C9ME00065H (* indicates equal contribution and co-first authorship)

1 **Jiale Shi**, Hythem Sidky, Jonathan K. Whitmer. "Novel Elastic Response in Twist-bend Nematic Models." ***Soft Matter*** 2019, 15**,** 8219-8226. (**inside front cover**) DOI: 10.1039/C9SM01395D

In Preparation

4 **Jiale Shi**, Qiang Cui. "Quantifying the Distance between Protein Dynamics with Graph Theory." *in preparation*.

3 **Jiale Shi**, "Conditional Generation of Block Copolymers with Large Language Models Fine-tuning." *in preparation.*

2 **Jiale Shi**, Nathan J. Rebello, Debra J. Audus, Bradley D. Olsen. "BigFingerprint: A Topology-aware Extended-Connectivity Fingerprint for Polymers." *in preparation*.

1 Katharina A. Fransen, Julia Casey, **Jiale Shi**, Natalie D. Mamrol, Gabrielle F. Godbille-Cardona, Jiarui Lu, Alex M. Zappi, Debra J. Audus, Bradley D. Olsen. "Predictive Modeling for Polyester Biodegradability: Insights from Augmenting Binary Co-polyester Data with a Terpolymer Library." submitted.

GRANT WRITING EXPERIENCE

2024 Contributor, EPA Research Grant Proposal. PI: Bradley Olsen.

2023 Contributor, NSF CDS&E-MSS Program Proposal. PI: Bradley Olsen.

2023 Finalist, Independent MIT ChemE Postdoc Research Grant.

TEACHING & MENTORING EXPERIENCE

Fall/2023 Mentoring undergraduate Jiarui (Sabrina) Lu through MIT UROP.

Lu was admitted to MIT-Wellesley Double Degree Program.

Spring/2023 Kaufman Teaching Certificate Program (KTCP), MIT.

Spring/2019 TA, ChemE Thermodynamics I, University of Notre Dame

Fall/2018 TA, Advanced ChemE Thermodynamics, University of Notre Dame

Spring/2018 TA, ChemE Thermodynamics I, University of Notre Dame

Fall/2017 TA, Science of Engineering Materials, University of Notre Dame

LEADERSHIP & PROFESSIONAL SERVICE

Reviewer Macromolecules,

2/2024 Session Chair, 2024 MaRDA Virtual Annual Meeting

11/2023 MESD Poster Judge, 2023 AIChE Annual Meeting

8/2023 PMSE Poster Judge, ACS Fall 2023

3/2023 Session Chair, APS March Meeting 2023

9/2022-8/2023 MIT ChemE Postdoc Advisory Board Leads

2018-2021 Graduate Student Participant, Midwest Integrated Center for Computational Materials (MICCoM)

9/2018-5/2019 Graduate Student Representative at Notre Dame Graduate Student Union (GSU), University of Notre Dame

9/2017-4/2019 Social Chair at Chemical and Biomolecular Engineering Graduate Student Organization (CBEGSO), University of Notre Dame

PRESENTATIONS

Oral Presentations

9 Faculty Candidate Onsite Interview at Department of Materials Science and Engineering, University of Wisconsin-Madison Feb 2025 (**Invited Talk**)

8 Faculty Candidate Onsite Interview at Department of Chemistry, University of Alabama April 2024 (**Invited Talk**)

7 ACS Fall 2023, San Francisco, CA, Earth Mover's Distance as a Metric for Calculating Pairwise Similarity of Polymer Ensembles, August 2023. **(ACS PMSE Invited Talk)**

6 ACS Fall 2023, San Francisco, CA, Quantifying Pairwise Similarity of Complex Polymers, August 2023. **(ACS POLY Big Data Award**)

5 MIT-NIST Joint-Postdoc Application Interview, Predicting Adhesive Free Energies of Polymer-Surface Interactions via Machine Learning and Transfer Learning, September 2021. (**Invited Talk**)

4 Geometry and Topology meet Data Analysis and Machine Learning (GTDAML 2021), Predicting Adhesive Free Energies of Polymer-Surface Interactions with Machine Learning, July 2021.

3 IDEA SLAM, Soft Matter Far from Equilibrium – CHESS 2030 Workshop, Predicting Adhesive Free Energies of Polymer-Surface Interactions with Machine Learning, June 2021.

2 Department Seminar 2020 Fall, Department of Chemical and Biomolecular Engineering, University of Notre Dame, Novel elastic response in twist-bend nematic models, October 2020. (**Invited Talk).**

1 17th International Conference on Ferroelectric Liquid Crystals, Boulder, CO, Novel Elastic Response in Twist-bend Nematic Models, August 2019.

Poster Presentations

4 2024 AIChE Annual Meeting, San Diego, CA, Meeting the Faculty Candidate Poster Session, October 2024.

3 20204 Polymer Physics Gordon Research Conference, July 2024.

2 2024 MaRDA Virtual Annual Meeting, February 2024.

1 6th Annual Notre Dame-Purdue Soft Matter & Polymers Symposium, West Lafayette, IN, Novel Elastic Response in Twist-bend Nematic Models, September 2019. **(Best Poster Award.)**

References:

Prof. Jonathan K. Whitmer (Ph.D. Advisor at University of Notre Dame)

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University of Notre Dame

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Website: https://engineering.nd.edu/faculty/jonathan-whitmer/

Prof. Bradley D. Olsen (Postdoctoral Advisor at MIT)

Department of Chemical Engineering

Massachusetts Institute of Technology

77 Massachusetts Avenue, 66-558a, Cambridge, MA 02139

Email: bdolsen@mit.edu Phone: (617)715-4548

Website: https://cheme.mit.edu/profile/bradley-d-olsen/

Dr. Debra J. Audus (co-Postdoctoral Advisor at MIT)

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